NEUROFUSION AI | FINAL CLINICAL DIAGNOSTIC REPORT

REPORT ID: REP-20261102-6377CA

Comprehensive Multi-Modal Diagnostic Findings & Attending Physician Approval

November 02, 2026 • 18:57 UTC

PATIENT INFORMATION	ATTENDING NEUROLOGIST	EXAMINATION DETAILS
ID: NF-PD-TEST-002 **Name:** Margaret Sullivan **Age/Gender:** 72 yrs / Female **Status:** Parkinsonian Movement Disorder	**Name:** Dr. Clara Rodriguez, MD **License:** MD-NEURO-77491 **Dept:** Neurology & Movement Disorders **Hospital:** Metropolitan Neurological Inst.	**Scan Date:** 2026-11-02 **Report Date:** 2026-11-02 **Modality:** Comprehensive Multimodal (Spiral + Voice + MRI + UPDRS) **Device/Protocol:** NeuroFusionAI Clinical Protocol v2.5

PRIMARY AI PREDICTION: PARKINSON

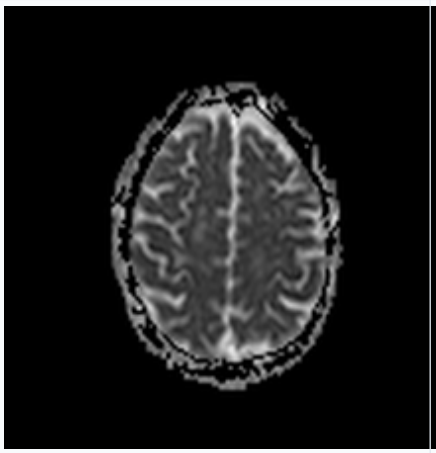
Confidence: 100.0%

Target: Parkinson's Disease

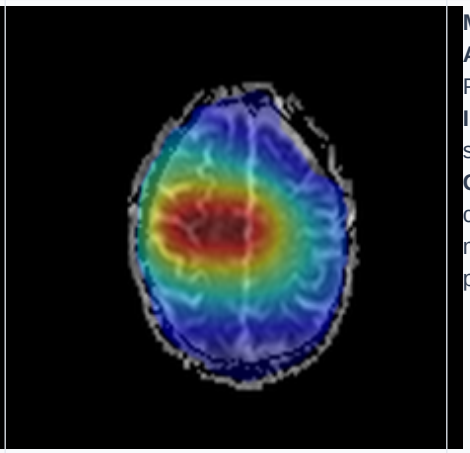
Class Probability Distribution: Parkinson: 100.0% • Normal Control: 0.0%

AI Confidence Calibrated via Temperature Scaling

1. MRI NEUROIMAGING & CONVOLUTIONAL GRAD-CAM EXPLAINABILITY



Input MRI Scan (Axial)



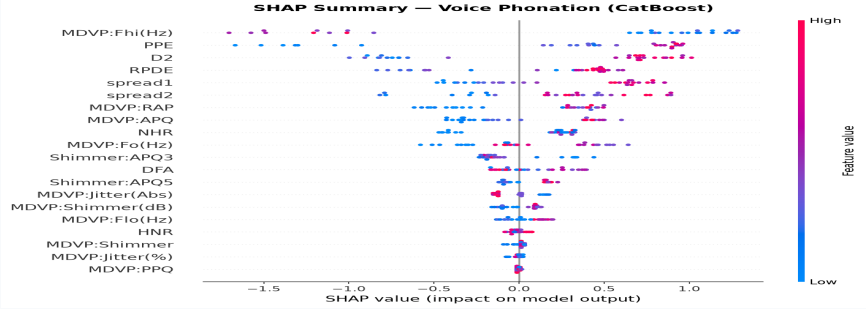
Grad-CAM Activation Overlay

Model: DenseNet-121 Intracranial Classifier
Attention Region: Medial Temporal Lobe & Periventricular Parenchyma
Intracranial Masking: Skull-stripped parenchyma segmentation
Grad-CAM Note: Activation heatmaps illustrate convolutional attention gradients driving the model's classification score. Not direct histological pathology.

XAI Interpretation

2. VOICE PHONATION ACOUSTIC ANALYSIS & SHAP FEATURE ATTRIBUTION

F0 Pitch (Hz)	Jitter (local %)	Shimmer (%)	HNR (dB)	Duration	Vocal Tremor
142.5 Hz	0.58%	3.92%	17.6 dB	4.5 s	**Detected**



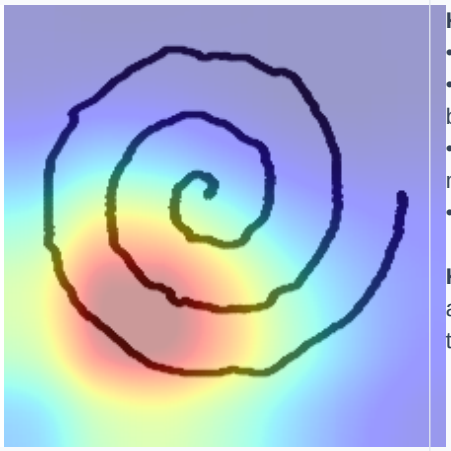
Acoustic Biomarker Interpretation: Elevated jitter (>0.38%) and shimmer (>2.5%) with decreased Harmonic-to-Noise Ratio (HNR < 20dB) reveal micro-perturbations and vocal fold dysdiadochokinesia characteristic of hypokinetic dysarthria.

SHAP Feature Attribution: Jitter local and MDVP:Fo(Hz) serve as dominant positive risk drivers.

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3. ARCHIMEDEAN SPIRAL MOTOR KINEMATICS & ATTENTION OVERLAY





Kinematic Parameters:
• Drawing Velocity: 24.8 mm/s
• Tremor Frequency: 5.4 Hz (4-6 Hz resting/kinetic band)
• Radial Velocity Variance: 0.42 (Elevated micro-tremor)
• Drawing Duration: 12.5 s

Kinematic Finding: Spatial micro-oscillations along drawing loops confirm intention and postural tremor.

Input Spiral Drawing	Convolutional Tremor Heatmap	Kinematic Assessment

4. TELEMONITORING CONTINUOUS MOTOR ASSESSMENT (MDS-UPDRS)

Predicted Motor UPDRS	Predicted Total UPDRS	Hoehn & Yahr Stage	Tremor/Rigidity Percentile	Monitoring Sessions
40.2 / 108	**50.4** / 176	**Stage 2.0** (Bilateral)	72th Percentile	1 Longitudinal Records
Clinical UPDRS Interpretation:	Predicted Motor UPDRS score indicates moderate bilateral parkinsonian motor impairment.			

6. MULTIMODAL CROSS-ATTENTION FUSION ARCHITECTURE

Diagnostic Modality	Feature Space Dimension	Cross-Attention Weight	Clinical Correlation
Volumetric Brain MRI	128 (DenseNet-121 Latent)	**30.0%**	Nigrostriatal & periventricular atrophy
Voice Acoustic Phonation	16 (Jitter, Shimmer, HNR, F0)	**25.0%**	Hypokinetic dysarthria micro-tremors
Spiral Drawing Kinematics	64 (CNN Kinematic Embeddings)	**25.0%**	4-6 Hz kinetic tremor oscillation
Telemonitoring UPDRS	18 (Continuous Motor UPDRS)	**20.0%**	Bilateral motor progression & rigidity

7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

RISK LEVEL:	Evidence-Based Diagnostic Recommendations:
HIGH CLINICAL RISK **Urgency:** Routine Specialist Review	• Initiate MDS-UPDRS structured clinical motor score evaluation. • Track resting tremor and bilateral bradykinesia progression. • Re-evaluate with longitudinal phonation acoustic tracking in 3 months.

ATTENDING PHYSICIAN OBSERVATIONS & NOTES:	CLINICAL APPROVAL STATUS
Progressive bilateral resting tremor, micrographia, postural instability, vocal hypophonia.	Signed by: *Dr. Clara Rodriguez, MD* License ID: MD-NEURO-77491 Status: **DIGITALLY APPROVED & CERTIFIED**

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REGULATORY MEDICAL DISCLAIMER & CLINICAL INTENDED USE:

NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.

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